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General Comment

See attached file.
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Attachments

AI action plan rfi 03.2025

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From:
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Subject:
A comment on the Development of an Artificial Intelligence (AI) Action Plan ("Plan")
in response to Request for Information (RFI)
issued by NSF on 02/06/2025, document # 2025-02305 (90 FR 9088)
that processes President Trump's Executive Order 14179 (**Removing Barriers to
American Leadership in Artificial Intelligence**).

Input:

**On the highest priority policy actions
that should be in the new AI Action Plan**

A suggestion about concrete AI policy actions that address the topics:
assurance of AI model outputs, risks, **regulation and governance**, technical and
safety standards, national security and defense, research and development,
innovation and competition, international collaboration, and import controls.

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government in developing the AI Action Plan and associated documents without attribution.

24th February 2025

1. Summary

- Securing American AI competitive edge is the correct primary objective. Thanks to Trump executive orders it is being reestablished.
- An omnipotent AI that is lacking safeguards is a weapon benefiting the enemy.
 - An overly potent AI lacking safety does help the enemy to burden the owner.
 - An overly restricted AI does help the enemy to maintain the upper hand.
 - The dichotomy “potency vs. safety” has merit, but was driven by a misinformed approach in Biden executive orders.
- AI safety and AI potency are not mutually exclusive.
 - The root cause of the fallacy thinking was potency and safety mapped on the same financial axes. The financial entanglement made the aspects development mutually competitive.
 - In fact, it is technically possible to strengthen the safety and boost AI potency at the same time, through the same expense. The development can be coherent.
 - AI safeguarding can come at nearly zero costs.
 - A set of organizational measures can disentangle the AI aspects and enable their coherent development.
- The proposed organizational measures to disentangle AI potency and AI safety have been tried and proven in many other industries.
 - A time-division principle is proposed to isolate the objectives.
 - A detailed policy set can be copy-pasted from the construction industry that regulates safety via building permits.
 - A governmental entity is required to establish and monitor the regulation.
 - The government entity wants to implement multiple separate bodies - at least one dependent on AI short-term growth and the other focusing on long-term safety.

2. Introduction

2.1 An ancient pattern

Once upon a time there was a monster in ancient Greek. It had untold powers that might bring immense destruction, or alternatively protect and crown princes. Kyriarchos (Κυρίαρχος) was adored and feared at the same time. Most people did not dare to come close to the forest, where Kyriarchos was living.

A group of young princes, dreaming of patriarchy, meant to domesticate the monster. Since no living thing ever survived a close encounter with the monster, nobody had any sense of what might be the best strategy for dealing with the monster. A prince named Evgenikos (Ευγενικός) proposed to bring treats regularly until the monster developed an addiction. Kyriarchos would then (supposedly) obey the master of the treats. Storgikos (Στοργικός) proposed to test the monster's taste first. He thought that focusing on a single bait will make the effort more affordable and effective. A prince named Synetos (Συνετός) proposed to test consumption too. He aimed for a reduced rationing to keep the monster under control, by restricting its ability to satisfy its desires.



It turned out the monster had been a nocturnal carnivore, who favoured hunting for fast moving mice. It had a ration comparable to that of a middle-sized domesticated cat. Kyriarchos was extremely cautious and sensible. It never ever went out of the forest just in case somebody was nearby watching. Over time the princes made themselves believe that they fed a cute cat. Yet Synetos insisted on regular testing. He noted soon that either the cat had been growing, or proliferating. The princes concluded, they would stop feeding if the consumption exceeded a certain threshold. Their hope was that an animal comparable to a bear could be tamed by three young athletes.

So they proceeded with their respective plans. All went well until one day the three, Evgenikos, Storgikos and Synetos, met their new pet. They now stood face to face with a three-story-building sized tiger. Kyriarchos was wise to cheat the experimenters out of their safety. It took the princes hostage demanding supplies in return for their release. The feeding continued until a complete impoverishment of their principedoms; Kyriarchos devoured stock, inventory and humans until nothing was left.

2.2 President Trump's right moves

In response to Biden-Harris [Executive Order 14110](#) of October 30, 2023 (Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence) [1] the largest AI developers and operators adopted a safety policy that red-teamed their AI models. Similar to the third prince's approach, our contemporaries want to undertake restrictive actions to contain the AI-threat below a threshold. It proved futile. The Greek princes were naive in measuring the non-human capabilities by human standards. **Trump [Executive Order 14148](#) of January 20, 2025 [2], rightly revokes Biden-Harris AI Executive Order 14110 [1].**

2.3 The issue yet to be addressed

Productivity requires the removal of barriers to American leadership [3]. Infrastructure vulnerability would be one of the barriers. Former approaches to deal with AI-vulnerability have been polarized and idealistic: the "monster" ought to be tamed, or effectively starved. Problem resolution is misguided by radical blueprints, falling to elusive targets and futile practices.

The issue presents a classic dilemma: an **unrestricted leverage of AI is outright dangerous, yet capping the leverage is counterproductive**. By nature, dilemmas demand balancing opposing forces, making them more challenging to address than standard problems. Relying on one-sided approaches often leads to detrimental consequences. Crafting an artificial Intelligence Action Plan could have been a constructive step toward tackling this issue effectively.

There does exist a solution for resolving the dilemma. This letter presents the key points. An operationalization of the solution would be a set of simpler tasks. The US government has a track record of resolving such tasks with great efficiency. The author abstains from those instructions until deliberately and specifically requested.

3. The problem

3.1 Key concept

The **essence of the problem lies in the leverage that an integrated AI can provide**. AI can evolve and elaborate any idea, as well as launch its implementation by a series of automated actions or clear instructions to a group of people. Implementation can happen at a scale that has never been seen before. The ability to scale is a positive development goal and the key objective in international competition.

A malicious user would use the leverage to further their goals, to detrimental results. A fully autonomous system can **execute harming orders with speed and magnitude that are not containable by** any known **law enforcement** system. This concerns the police, the defence department, and market regulators, as well. Even a tiny misdemeanor could be amplified and multiplied to awful proportions, dwarfing the most egregious felonies with respect to their impact. APTs could use the amplification against American interests.

3.2 Two-sided interventions

The dilemma “an unrestricted leverage of AI is outright dangerous, yet capping the leverage is counterproductive” emanates from two sides: 1) AI capability growth, 2) utility governance. Trump's Executive Order 14179 [3] aims to place the former back onto the industry agendas. However, that in itself would not be sufficient. The AI Action Plan must address the latter as well. For any solution to work effectively both sides have to be balanced at any given time.

The balance is existential, it falls out of the economic realm. The market has to mature before the balance (a stale ecosystem) is maintained by the market players. It is initially a task for the federal institutions to establish proper AI governance at par with the AI growing capacity. Should it be neglected, a disaster of epic proportions would be addressed by the question “when”. The adversaries will certainly stress the situation and use the unattended AI powers to command American failure. Accelerated development would require intensive government interventions on both sides of the dilemma.

The **concept of balance has not been taken into consideration** so far. This could be the most important feature in the upcoming AI Action Plan and legislation set. A similar approach once made Napoleon famous, as he introduced the Napoleonic Code with balanced rights and responsibilities.

3.3 One-sided scenarios are failures

Let's consider an example of one-sided intervention. Reiterating the previous Biden-Harris AI safety approach [1], the AI players were essentially set to control the action amplification factor. Red-teaming was to test how much a Kyriarchos can chew at once, i.e. how much harm AI is capable of causing. If it were too much, the AI-model was supposed to be held back following the Synetos approach.

This approach has not separated the two sides of the dilemma. As a result, every safety effort was mapped onto the commercial axes, forming a negative feedback mechanism. It is

antithetical to the AI arms race, which is currently unfolding and gaining steam. Companies are vying for commercial success, which lets them acquire enough resources to allow them to stay in the race. The negative feedback loop puts a contender back in the race, while others proceed. It would be naive and overly optimistic to anticipate its fair implementation. AI companies are predisposed to yammer about more subsidies for increasingly deceptive and inefficient safety activity.

Tying safety to the action amplification factor, or expressed differently, setting an AI-leverage cap, is a shot in the leg, too. The amplification factor will grow and exceed any threshold as a result of the dynamics of the international contest. The NSF and NITRD National Coordination Office would also want to help the US economy to outcompete any contender for AI capabilities for as long as it goes.

If we negate the Synetos approach, i.e. apply positive feedback, it will also fail. By getting rid of testing, we do not eliminate the factor that constrains growth. We **replace a deliberate decision by a random trigger** that would painfully constrain the proverbial monster and provoke it to strike unexpectedly. Allowing a random trigger to initiate the chain of events as it was in the Greek story, is extremely reckless. Unfortunately, such is the situation in which we find ourselves.

There are multiple parties, who directly benefit from vibrant AI development. All of them accelerate the market dynamics. If one of the parties, let it be for example the energy supply party, ceases their participation for legitimate reasons, or even at a whim, the market pull will first empty out all dependents. That is, the energy shortage will trigger uneven and exorbitant prices. Retail consumers wouldn't be capable of competing with bulk purchasers, like AI-computing farms. AI-farms would be overpaying, exploiting their productivity drivers and higher margins. Regulation, by itself, will not be able to cope with the phenomenon. Any preference for little Jill would just trigger spreading the AI capacity onto the edges, similar to how mainframes were dissolved by the on-premise PC-like servers. In the end, it would be even harder to balance distributed AI deployment and to control its impact.

While spreading was a desirable feature in the PC-era, contributing to the software availability and infrastructure (Internet) resiliency, now it is neither beneficial nor desirable. A consolidated AI development and deployment would be cheaper to sponsor, easier to oversee, simpler to govern, and quicker to saturate the market, thus quicker producing a stale self-regulated ecosystem and a responsible AI culture. Software quality was optional during the PC-era. It evolved over time. The AI-era demands certain quality standards (benevolence) right upfront.

4. Solution

4.1 Requirements

A sponsored AI development would benefit from a **balanced approach**. Such an approach encompasses a) **AI capacity** growth that is **matched by** 2) an appropriate **governance standard**. These objectives should be isolated and driven separately, offering no opportunity for cross-mapping, merging, and detrimentally competing.

A **federal agency** must shape balanced AI development, as well as monitor the AI operations to **ensure that practice matches the intention** (documentation).

The key principles here are:

- A balance between AI capability and AI control has to be an
 - auditable and
 - audited feature.
- The balance method is not fixed. The balance utility must be checked if the AI control:
 - is capable of containing the full leverage scale, or alternatively the spillovers are properly insured,
 - has a reasonable safety margin for the anticipated leverage,
 - strength is evaluated in relation to the maximal anticipated AI-power, i.e. the controls are dimensioned not by the mathematical expectations but by the top outliers;
- AI control does not weaken and corrupt over time.
 - Evaluations do not draw measures from the human bases, but from the (theoretical) technical top values,
 - the engineering teams work in a framework that is insulated against malfeasance,
 - there is an approved mitigation action plan to address the contingency of an inadvertently triggered event.

It has to be assumed that despite all the efforts, there would be no complete oversight. At the AI-capabilities exceeding that of humans it is theoretically not feasible. Therefore the governance standards should employ a self-adjustment mechanism. Self-adjustment would augment human oversight. Assuming that AI capacity can substantially grow under the radar, the mechanism is to demand at any facility disregard its actual AI-capacity and staff.

4.2. Policy actions

From the standpoint of industrial investment, a two-legged development is much too expensive. Therefore any attempt to mandate it, in the absence of rigorous enforcement, would provoke fraud. The second leg reserved for sustainability, robustness and balance is an outright executive's heresy, who spend life creating and exploiting imbalances. An imbalance is in the blood of the market competitors. The balancing and enforcement job is specifically reserved for a governance entity.

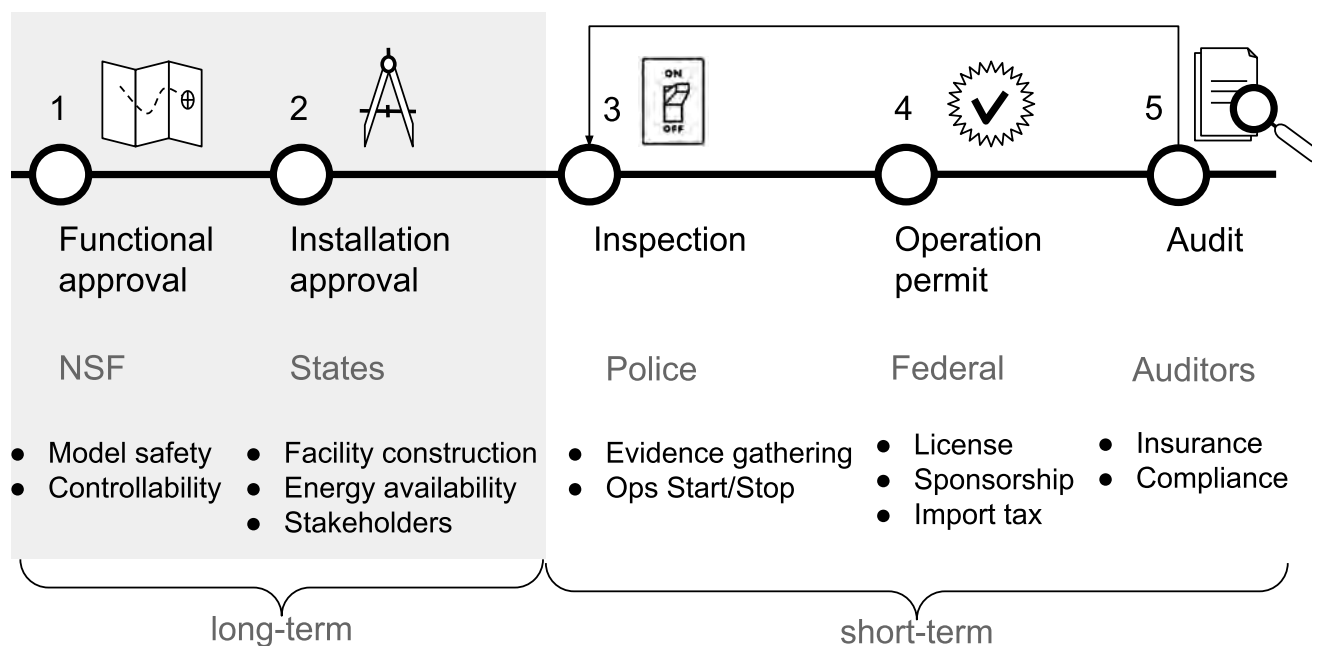
The sheer scale of the AI phenomenon will draw resources noticeable on the World GDP scale. Even the most formidable countries would experience challenges facing the economic pressures. The governance entity in charge of AI regulation should be not less than federal, if not pan-regional in its scope.

A **method to isolate the dilemma objectives** clearly may be **time-division**. The reaction time and pace in the control loops could be drastically different. Then a cross-mapping becomes impractical, reducing the chances that the system degrades into a one-sided problem.

For example, utility governance could be driven by the long-term incentives exclusively, and the AI capability growth by short-term. The market institutions could take the short-term regulations over. A special purpose non-profit entity could regulate the long-term. All short-termism should be prohibited by its mandate. A coordination across the entities for synergetic effect could be mandated to additional parties, thus prohibiting migration of the focus. Such a governance model is proposed as follows.

The federal government establishes **two entities**. The entities **approve and affirm AI licenses**. One entity checks the documentation (long-term objectives). Another entity orders regular auditing of the AI operations (short-term objectives). A **third entity** might get a **law enforcement** mandate to swiftly dismantle rogue AI operations.

The AI licensing process could resemble a construction permit process. The construction industry also places long-term safety in the forefront. One could tap those experiences and legislative solutions that already exist today.



4.2.1 Approval

The prince's failure was to assess the monster by human originated units of measurement. It led to a systematic underestimation. This was a direct consequence of their

practical/commercial approach. Easily obtained facts take precedence over hard to get hypotheses. The same bias is to be anticipated in any for-profit organisation.

A theoretical approach would have drawn on the secondary data sources, like paw size, ground displacement, concealed monitoring equipment, etc. It could provide an alternative measurement unit base and a plausible estimation of the monster size and strength. Thus the prince's desire to test and tame the monster was fundamentally correct, but poorly implemented due to a biased perception.

An entity, which is not being pressured by a commercial agenda, would not cut corners on AI-safety investigation methods. This approach has been implemented many times over. **NSF** could be a useful intermediary that approves AI operation licenses. The approval has to be based on the theoretical controllability. The decision underpinning will be provided by the companies. NSF would need to prove the evidence and logic. More importantly, a party like NSF could also draw conclusions based on the *state of the art* knowledge from multiple independent sources, which otherwise would not be shared between competing parties.

4.2.2 Affirmation

An auditor periodically affirms that the approved and the operated systems are identical.

Financial auditors could expand their role to include monitoring AI operations.

The federal government may then gate incentive funds distribution on the basis of the compliance results.

4.2.3 Enforcement

Despite a rigorous licensing process, there could be regular demand for an unprepared, unannounced inspection, as well as for immediate operational interventions. A law enforcement unit ought to react swiftly to the crime, which is amplified by AI.

A fair share of petty crime would be effectively held in check by the unit. To expand the checked area, the unit would be differentiated by AI preparedness and a mandate to preemptive strike without a court order and public exposure on certain occasions. The law enforcement unit ought to establish special weapons and tactics that help to operate in a cyber environment.

Governments at all levels may establish a feedback mechanism that allows unobtrusive help by the industry players. The players would contribute to structuring, as well as sponsor the centralized **dedicated law enforcement unit**, which has **higher reaction speed** and outreach. The market player's incentive should be to offload the tasks onto government agencies, and to elicit economies of scale. Multiple players would contribute to the same shared agency. Alternatively each player can take responsibility for the damage incurred by its leveraged attacks, i.e. if they were performed in its facility. Over time, as the AI market matures, the enforcement unit could be privatized and run in the interest of the players to maintain the status quo.

4.3 Solution examples

It is expected that the AI operators will deny and actively oppose some requirements. The self-adjustment and malfeasance protection frameworks being likely examples. Yet it is of primary importance that the requirements remain fulfilled, otherwise a commercial suboptimization will disable any balanced and capable governance.

In order to sweep off the argument that those requirements are not realistic, here are references to the methodologies that do exactly what is required. The examples are not meant to limit readers' thinking. There could be other approaches that fit the demand.

4.3.1 Self-adjusting AI control

AI system leverage is the primary reason to deploy AI. The leverage develops along a growing AI system scale today. The AI scale grows at breathtaking speed. The AI decision making scale would exceed that of the transnational corporations in a matter of a few years. No individual, or group, would be able to compete with AI in the near future. AI systems themselves and their operators separately will be predisposed to cheat without being noticed from that point onwards. An overscaled oversight would not carry us much further either.

All AI systems with a potential to grow and get more sophisticated over time must adopt some mechanisms for self-adjustment and self-alignment. The mechanism would embed a self-induced “moral” and “ethics” into the AI models. The mechanism is similar to what we vote for and encounter in legislation drafters. It will likely become better with AI capacity growth. Hence, as people loosen their grip, AI gains through self-improvement. An option to implement such a mechanism is described in [4].

4.3.2 Malfeasance framework

No technical system is immune to the threat of treachery or incompetence from service personnel. Covert changes in system setup is a long-known problem in banking. The banking frameworks are a step in the right direction, but are still far from being water-tight. A significantly expensive bank crime would trigger a significantly expensive chase. The trick is to raise the bar for a minimal possible crime up to the point that a successful chase is inevitable.

Such an approach may not work for AI. Whoever gains full control over AI leverage will be in the position to outcompete any effort by detectives and prosecutors to catch up.

An AI framework has to shift focus on not letting malign individuals infiltrate and settle in any AI control center. An explanation of social dynamics and instructions for such a framework are described in [5].

References

- [1] Biden, J., Executive order 14110, “Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence”, [FedReg](#), Oct. 30th, 2023.
- [2] Trump, D., Executive order 14148, “Initial Rescissions of Harmful Executive Orders and Actions”, [FedReg](#), Jan. 20th, 2025.
- [3] Trump, D., Executive order 14179, “Removing Barriers to American Leadership in Artificial Intelligence”, [FedReg](#), Jan. 23rd, 2025.
- [4] Spasokukotskiy, K., “Synthetic consciousness architecture”, [Front. Robot. AI](#), 11.2024.
- [5] Hardy, J.B., “The hidden game revealed: The game & the players”, iUniverse, 2010.